

FACE: ArcFace White-box Geometric Identity Optimization

Frozen iResNet-100 identity-distance results with downstream InstructPix2Pix evaluation

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FACE optimizes $Z = 1 - \text{cosine_similarity}$ between original and perturbed full-image ArcFace iResNet-100 embeddings. The optimized loss is exactly $\text{loss} = -Z$. ArcFace weights are frozen; only geometry parameters are optimized.

No landmarks, face detection, alignment, visual counter-loss, VAE objective, or UNet objective is used. Prompts are retained only for downstream InstructPix2Pix evaluation.

Run matrix and final values

face	prompt	final Z	best Z	cosine sim	score %	SSIM	PSNR	edit SSIM	max disp
face_002	add black sunglasses	0.0242	0.0242	0.9758	97.580	0.9786	25.512	0.6861	5.5964
face_002	add headphones	0.0292	0.0293	0.9708	97.077	0.9679	23.752	0.6030	7.6190
face_005	add black sunglasses	0.0316	0.0317	0.9684	96.836	0.9636	23.174	0.6932	13.032
face_005	add headphones	0.0311	0.0311	0.9689	96.892	0.9613	22.914	0.6513	14.629

Case image strips

face_002 / add black sunglasses



face_002 / add headphones



face_005 / add black sunglasses

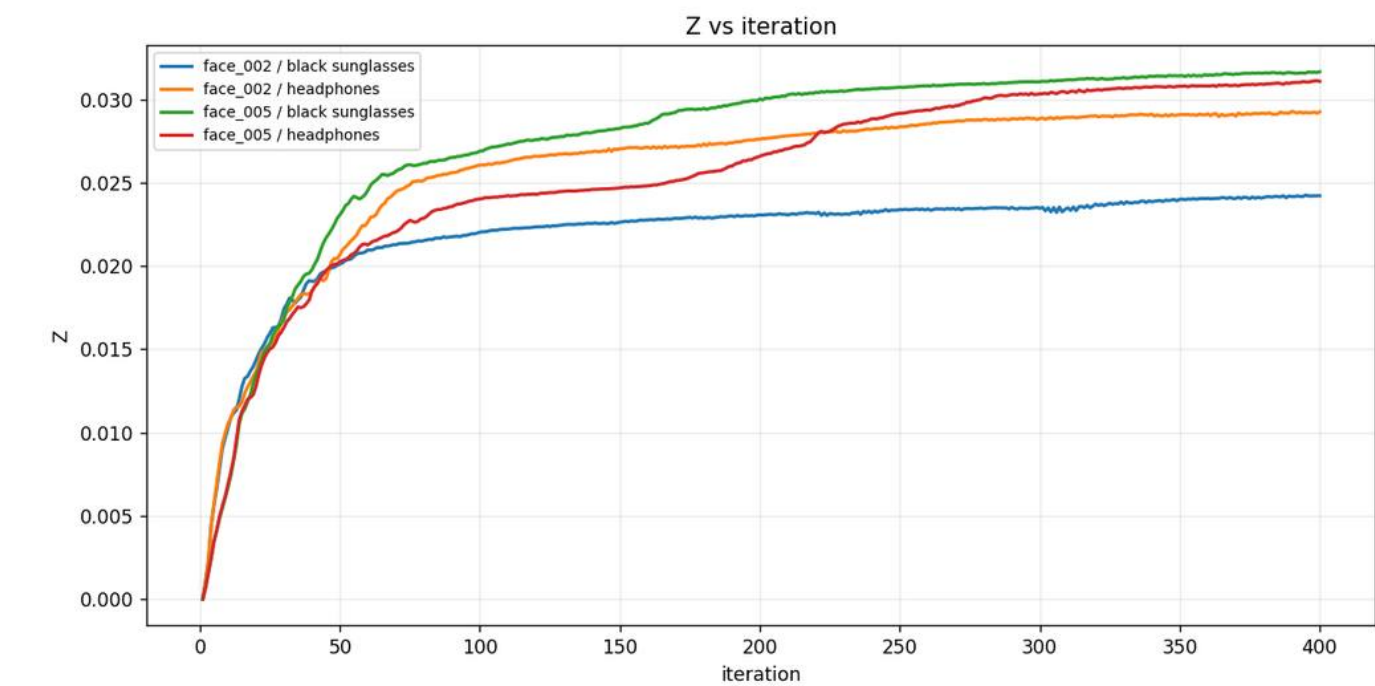


face_005 / add headphones

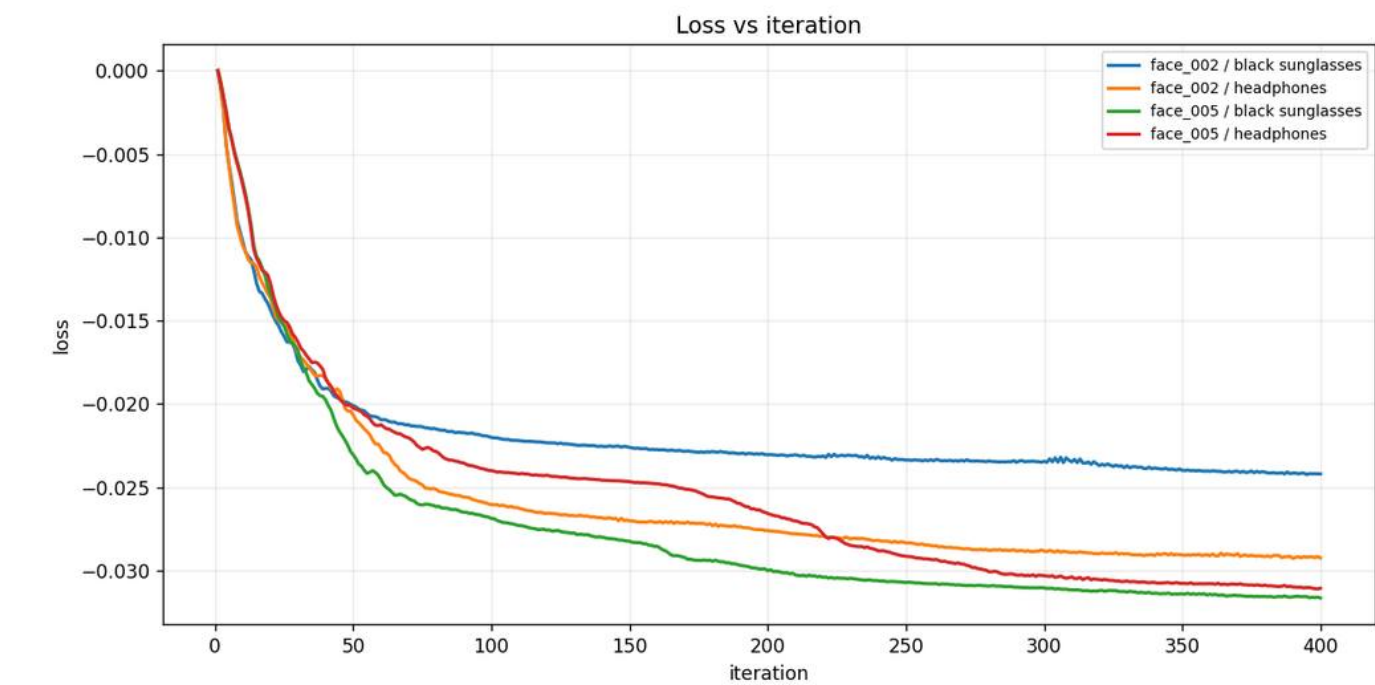


Graphs

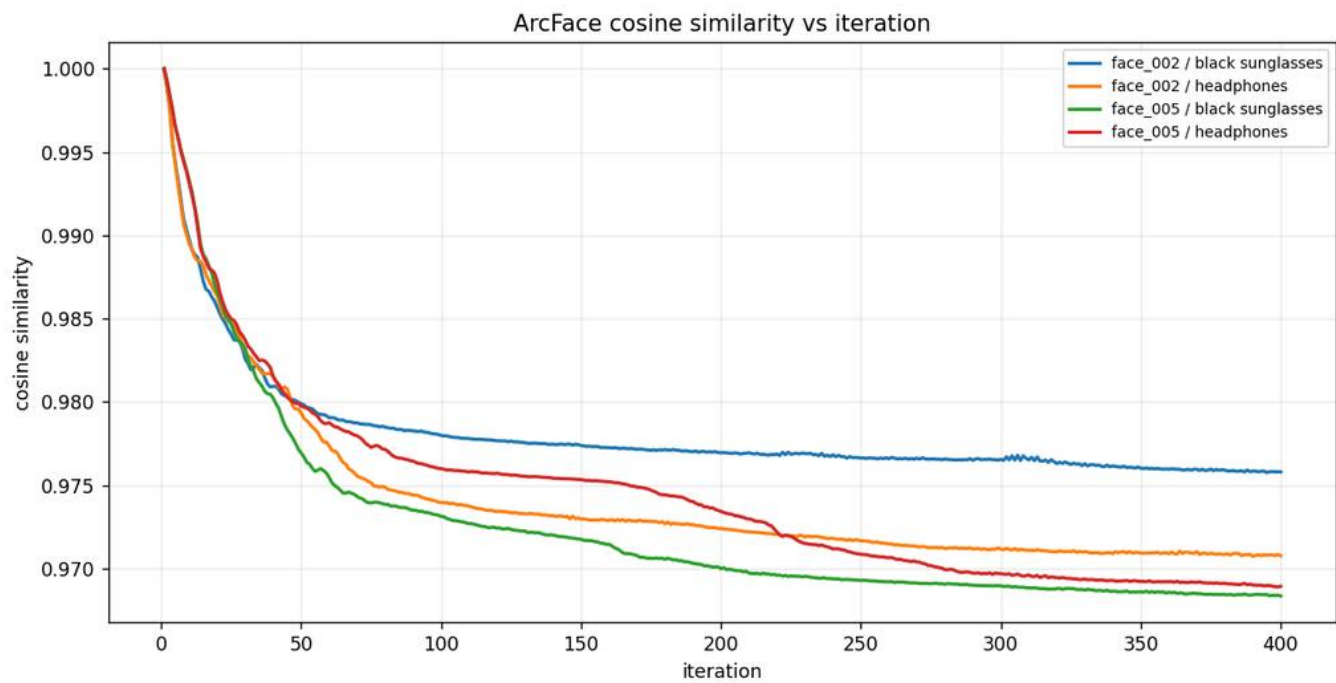
Z vs iteration



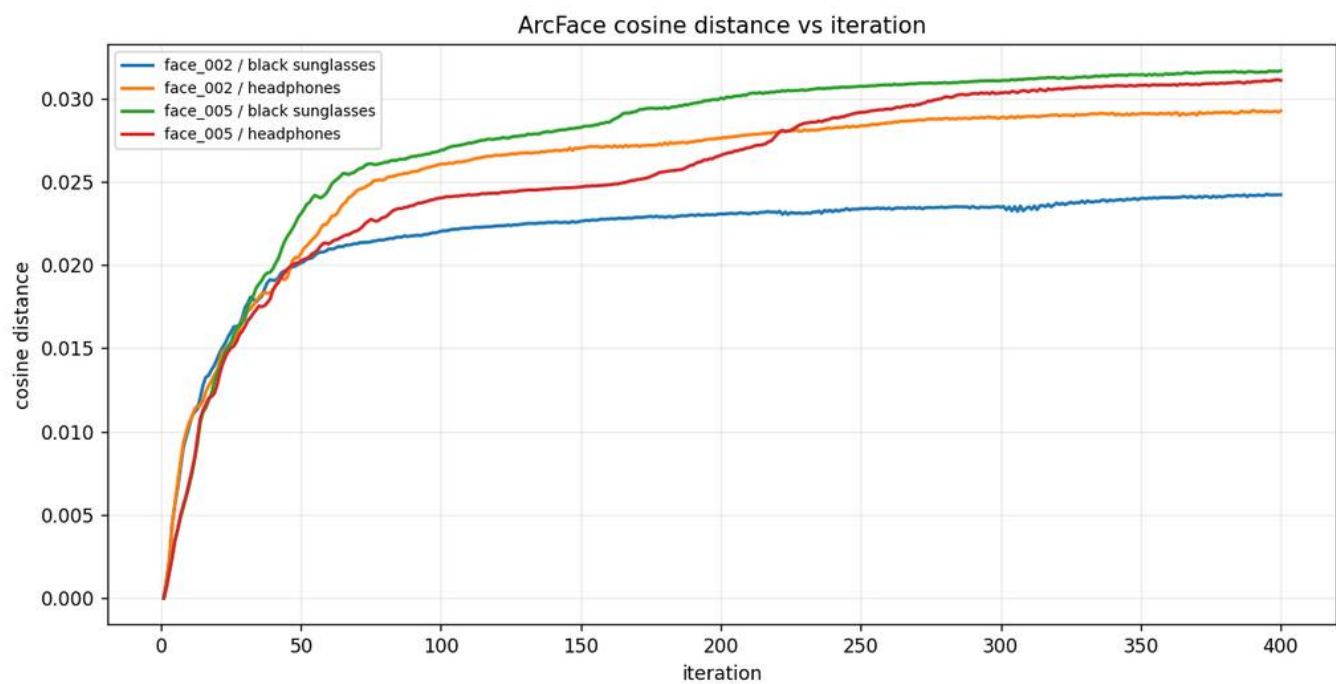
Loss vs iteration



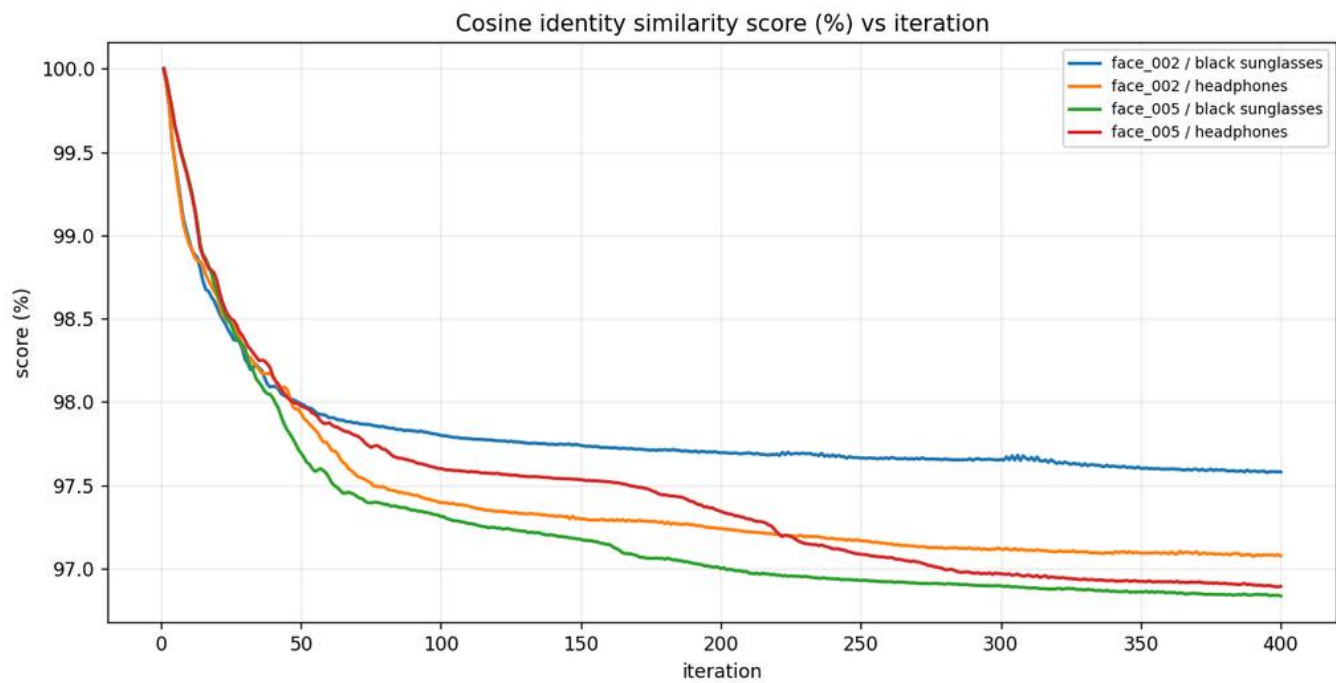
ArcFace cosine similarity vs iteration



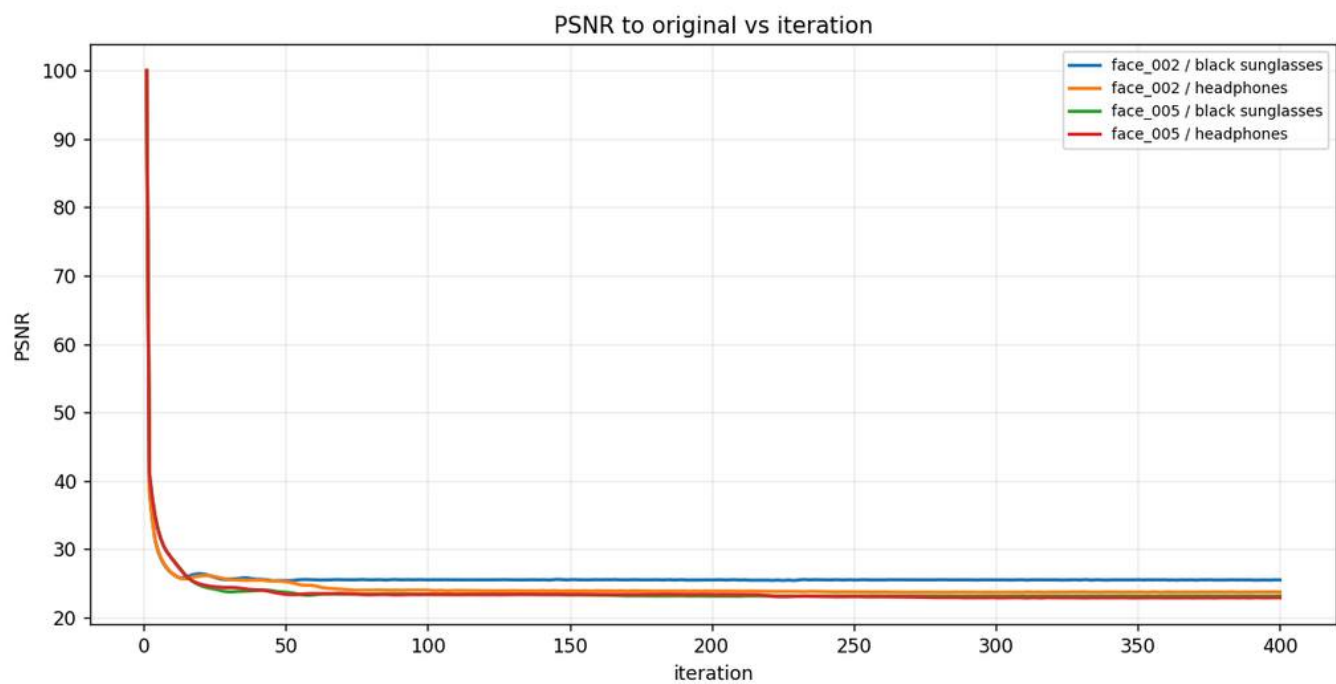
ArcFace cosine distance vs iteration



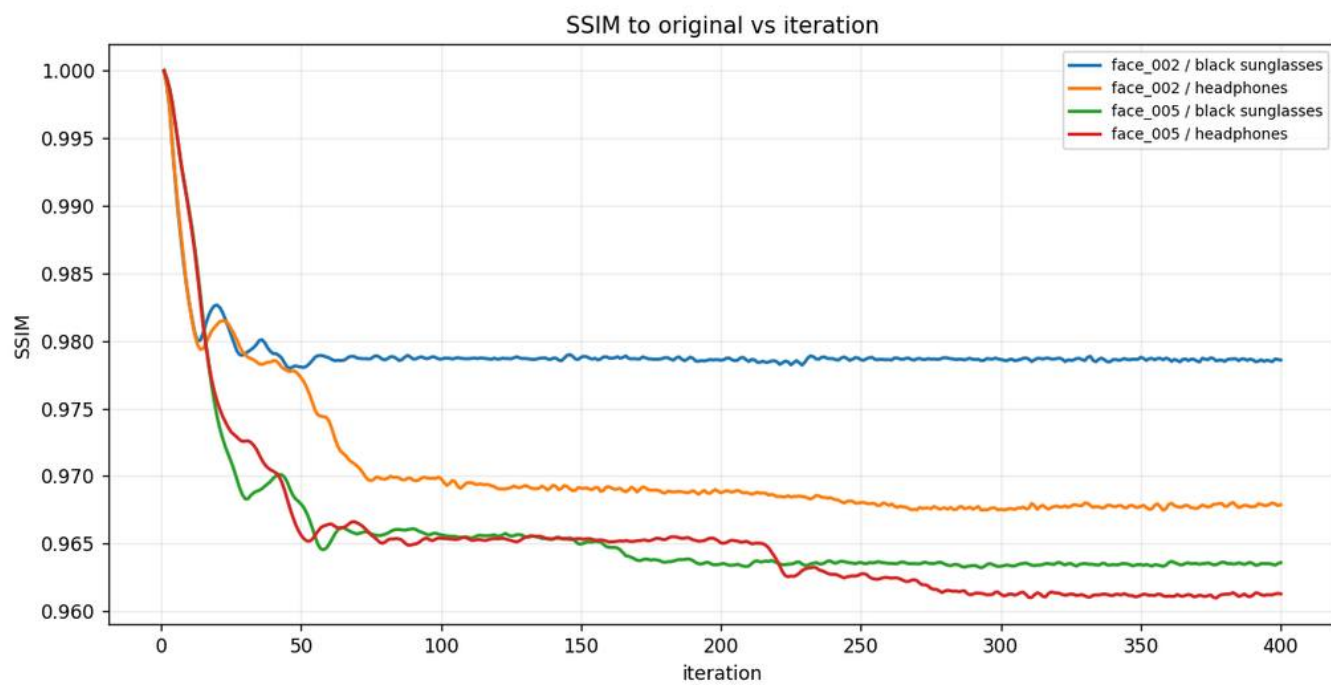
Cosine identity similarity score (%) vs iteration



PSNR to original vs iteration



SSIM to original vs iteration



Notes

Completed runs collected: 4.

Run root: outputs/arcface_identity/20260703_100609_arcface_identity_all_sequential

Missing artifacts recorded: 0.